With data, fibre entry is equivalent to

Consider the three rays in turn.

:

Can the refracted ray enter the fibre?

Using , ; hence

Substitute into :

Substitute into :

Since

the ray enters the fibre.

Can total internal reflection occur at the core - cladding interface? Let the critical angle be , then

By , total reflection requires to satisfy the following:

Now meets ; thus it propagates by total internal reflection in the fibre.

:

Using :

: ray cannot enter the fibre right of .

:

Using :

: ray enters the fibre right of .

Then

Negative: backward ray meets the axis left of , hitting the upper boundary from below (Fig.). ; it can propagate in the fibre by total internal reflection.

Fig.